

# ACHINTYA RAI

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## EDUCATION

### University of California, San Diego

Sept. 2024 – June 2028

*Bachelor of Science in Computer Engineering*

**GPA: 3.9/4.0**

- Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow

## WORK EXPERIENCE

### Adobe

San Jose, CA

*Software Engineering Intern*

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, identifying **86,387** duplicate records and a **47%** null rate to establish the primary data source for production monitoring.
- Designed a time-series anomaly detector iterating over statistical algorithms to handle multi-layer seasonality across **10+** workflows and **30+** countries, shipping code to production.
- Developed SQL and MLOps pipelines for weekly retraining and hourly inference; identified a **26-hour** processing outage across **1,272** segments as the primary validation benchmark.

### VALT Health

Remote

*Founding Software Engineer*

Aug. 2025 – Feb. 2026

- Engineered a Flask backend handling **100k+** datapoints from fitness devices (Fitbit, Apple Watch), architecting well-structured schemas to unify data formats across **20+** sources.
- Implemented asynchronous batch processing in GCP, achieving **35%** faster execution and **60%** fewer redundant writes, improving the performance and reliability of the system.

### Decklar (formerly Roambee)

Sunnyvale, CA

*Software Engineering Intern*

June 2024 – Aug. 2024

- Optimized Python and NumPy pipelines for **Amazon, Coca-Cola, and Samsung**, achieving **15%** higher throughput and **20%** faster reporting across **1M+** weekly events.

## RESEARCH

### Graph Theory & Neural Architecture

San Diego, CA

*Undergraduate Research Fellow (NSF REU)*

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), applying custom algorithms to compress **500GB** checkpoints to **13GB** localized manifolds.
- Optimized preprocessing scripts to achieve a **30x** runtime reduction (30 mins to <1 min), enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.

### Data Science Teaching & Learning Lab

San Diego, CA

*Undergraduate Researcher*

Sept. 2024 – Present

- Architected a multi-agent AI tutoring framework with hallucination-reduction strategies, achieving **95%+** accuracy and processing **100k+** data points via high-throughput agentic workflows.
- Developed an LLM-augmented labeling tool using the Gemini API, automating annotation of **16k+** data points and reducing labeling time by **90%** through iterative refinements.

## PROJECTS

### NanoGPT | Python, PyTorch, C++ |

Mar. 2026

- Built a GPT-style transformer in **Python/PyTorch** with multi-headed attention and backpropagation, deepening computer science fundamentals from scratch.
- Compiled sub-modules in **C++** to cut per-token latency by **25%**; engineered dataloaders sustaining **100%** GPU utilization on resource-constrained hardware.

### SquaredUpChess | Java, React, Django, WebSockets, AWS

Aug.–Dec. 2025

- Architected **20+ Django REST API endpoints** for auth and payments on **AWS**; established **CI/CD** pipeline ensuring **99.9%** deployment reliability.
- Optimized **SQLite** via strategic indexing and implemented real-time sync with **WebSockets** and **React** for low-latency multi-user scalability.

## TECHNICAL SKILLS

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**Languages:** C++, Python, Java, TypeScript, SQL (Postgres/SQLite), JavaScript, Bash, HTML/CSS

**Development:** Django, React, Flask, FastAPI, Spring Boot, Node.js, Asyncio, WebSockets, PyTest, JUnit

**AI & Cloud:** PyTorch, AWS, GCP, OpenAI/Gemini API, Docker, GitHub Actions, Linux